Homework 1

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Problem 1

(a)

We show $(A \cap C) \cup (B \cap C) \subset (A \cup B) \cap C$.

Let $x \in (A \cap C) \cup (B \cap C)$. Then either $x \in A \cap C$ or $x \in B \cap C$.

If $x \in A \cap C$, then $x \in A$ and $x \in C$, so $x \in A \cup B$ and $x \in C$, hence $x \in (A \cup B) \cap C$.

If $x \in B \cap C$, then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in C$, hence $x \in (A \cup B) \cap C$.

Thus $(A \cap C) \cup (B \cap C) \subset (A \cup B) \cap C$. Combined with the inclusion proved in class, equality holds.

(b)

From class, A^c and B are independent, so $P(A^c \cap B) = P(A^c)P(B)$. Note that $A^c = (A^c \cap B) \cup (A^c \cap B^c)$ is a disjoint union, hence

$$P(A^c) = P(A^c \cap B) + P(A^c \cap B^c).$$

Therefore

$$P(A^c \cap B^c) = P(A^c) - P(A^c \cap B) = P(A^c) - P(A^c)P(B) = P(A^c)\big(1 - P(B)\big) = P(A^c)P(B^c).$$

Hence A^c and B^c are independent.

Problem 2

(a)

Note that $1_{\{N \geq 1\}} \leq N$, since both sides are nonnegative integers and the inequality holds for every realization of N. Taking expectations,

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) = E\big[1_{\{N \geq 1\}}\big] \leq E[N] = E\Big[\sum_{i=1}^{20} 1_{A_i}\Big] = \sum_{i=1}^{20} P(A_i).$$

With $P(A_i) = 0.01$,

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) \leq 20 \cdot 0.01 = 0.2.$$

(b)

For every nonnegative integer n,

$$1_{\{n\geq 1\}} \geq n - \binom{n}{2},$$

since the right-hand side equals 1 for $n \in \{1, 2\}$ and is at most 0 for $n = 0$ or $n \geq 3$. Applying this to N and taking expectations,

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) = E\big[1_{\{N\geq 1\}}\big] \geq E[N] - E\Big[\binom{N}{2}\Big].$$

Expanding,

$$\binom{N}{2} = \sum_{1\leq i<j\leq 20} 1_{A_i\cap A_j}, \qquad E\Big[\binom{N}{2}\Big] = \sum_{i<j} P(A_i \cap A_j),$$

so

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) \geq \sum_{i} P(A_i) - \sum_{i<j} P(A_i \cap A_j).$$

Under pairwise independence, $P(A_i \cap A_j) = P(A_i)P(A_j) = 10^{-4}$. There are $\binom{20}{2} = 190$ pairs, hence

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) \geq 0.2 - 190 \cdot 10^{-4} = 0.2 - 0.019 = 0.181.$$

(c)

Under mutual independence,

$$P\Big(\bigcup_{i=1}^{20} A_i\Big) = 1 - \prod_{i=1}^{20}\big(1 - P(A_i)\big) = 1 - (0.99)^{20} \approx 0.1821.$$

This lies between the lower bound 0.181 and the upper bound 0.2.

Pairwise independence alone does not determine $P(\bigcap_i A_i^c)$, because that probability depends on the full joint distribution of $(1_{A_1}, \ldots, 1_{A_{20}})$, not merely on the two-dimensional margins. Mutual independence is what factors the product $\prod(1 - P(A_i))$.

Problem 3

Let $S \in \{\text{dot}, \text{dash}\}$ denote the symbol sent and R the symbol received. Then

$$P(S = \text{dot}) = \tfrac{3}{7}, \qquad P(S = \text{dash}) = \tfrac{4}{7},$$

and

$$P(R = \text{dash} \mid S = \text{dot}) = \tfrac{1}{4}, \qquad P(R = \text{dot} \mid S = \text{dot}) = \tfrac{3}{4},$$
$$P(R = \text{dot} \mid S = \text{dash}) = \tfrac{1}{3}, \qquad P(R = \text{dash} \mid S = \text{dash}) = \tfrac{2}{3}.$$

(a)

By Bayes,

$$P(R = \text{dash}) = \tfrac{1}{4} \cdot \tfrac{3}{7} + \tfrac{2}{3} \cdot \tfrac{4}{7} = \tfrac{3}{28} + \tfrac{8}{21} = \tfrac{9}{84} + \tfrac{32}{84} = \tfrac{41}{84},$$

so

$$P(S = \text{dash} \mid R = \text{dash}) = \frac{\frac{2}{3} \cdot \frac{4}{7}}{\frac{41}{84}} = \frac{8/21}{41/84} = \frac{32}{41}.$$

(b)

Write d = dot and D = dash. The four possible sent messages for a received message dd are dd, dD, Dd, and DD. Under independence across positions,

$$\begin{aligned} P(S = \text{dd}) &= \left(\tfrac{3}{7}\right)^2 = \tfrac{9}{49}, & P(R = \text{dd} \mid S = \text{dd}) &= \left(\tfrac{3}{4}\right)^2 = \tfrac{9}{16}, \\ P(S = \text{dD}) &= \tfrac{3}{7} \cdot \tfrac{4}{7} = \tfrac{12}{49}, & P(R = \text{dd} \mid S = \text{dD}) &= \tfrac{3}{4} \cdot \tfrac{1}{3} = \tfrac{1}{4}, \\ P(S = \text{Dd}) &= \tfrac{4}{7} \cdot \tfrac{3}{7} = \tfrac{12}{49}, & P(R = \text{dd} \mid S = \text{Dd}) &= \tfrac{1}{3} \cdot \tfrac{3}{4} = \tfrac{1}{4}, \\ P(S = \text{DD}) &= \left(\tfrac{4}{7}\right)^2 = \tfrac{16}{49}, & P(R = \text{dd} \mid S = \text{DD}) &= \left(\tfrac{1}{3}\right)^2 = \tfrac{1}{9}. \end{aligned}$$

The unnormalized posteriors $P(R = \text{dd} \mid S)\, P(S)$ are

$$\frac{9}{16} \cdot \frac{9}{49} = \frac{81}{784}, \quad \frac{1}{4} \cdot \frac{12}{49} = \frac{3}{49} = \frac{48}{784}, \quad \frac{48}{784}, \quad \frac{1}{9} \cdot \frac{16}{49} = \frac{16}{441} = \frac{256}{7056}.$$

Clearing a common denominator $7056 = 9 \cdot 784$,

$$\frac{729}{7056}, \quad \frac{432}{7056}, \quad \frac{432}{7056}, \quad \frac{256}{7056}.$$

The normalizing constant is $729 + 432 + 432 + 256 = 1849$, so

$$\begin{aligned} P(S = \text{dd} \mid R = \text{dd}) &= \frac{729}{1849}, & P(S = \text{dD} \mid R = \text{dd}) &= \frac{432}{1849}, \\ P(S = \text{Dd} \mid R = \text{dd}) &= \frac{432}{1849}, & P(S = \text{DD} \mid R = \text{dd}) &= \frac{256}{1849}. \end{aligned}$$

Problem 4

(a)

For $x \in [0, 1)$, $x^n \to 0$; at $x = 1$, $h_n(1) = 1$ for all n. Thus $h_n(x) \to 0$ for Lebesgue-almost every $x \in \mathbb{R}$ (the single point $\{1\}$ has measure zero). Moreover $|h_n(x)| \le 1_{\{[0,1]\}}(x) =: g(x)$, and $\int g < \infty$. By the Dominated Convergence Theorem,

$$\lim_{n \to \infty} \int_{\mathbb{R}} h_n(x)\, dx = \int_{\mathbb{R}} 0\, dx = 0.$$

(Directly: $\int_0^1 x^n\, dx = 1/(n+1) \to 0$.)

(b)

Let (t_n) be any sequence with $t_n \to t_0$. For n large, $|h(x,t_n)| \le g(x)$ with g integrable, and $h(x,t_n) \to h(x,t_0)$ pointwise. DCT gives

$$\lim_{n\to\infty} \int h(x,t_n)\,dx = \int h(x,t_0)\,dx.$$

Since the limit along every sequence $t_n \to t_0$ equals $\int h(x,t_0)\,dx$, we obtain

$$\lim_{t\to t_0} \int h(x,t)\,dx = \int h(x,t_0)\,dx.$$

(c)

(i) Factor out $e^{t_0 x}$:

$$q_h(x) = e^{t_0 x} \cdot \frac{e^{hx}-1}{h}.$$

Set $u = hx$. As $h \to 0$ we have $u \to 0$, and

$$\frac{e^{hx}-1}{h} = x \cdot \frac{e^u - 1}{u} \to x \cdot 1 = x,$$

using the standard limit $\lim_{u\to 0}(e^u-1)/u = 1$. Therefore

$$q_h(x) \to e^{t_0 x} \cdot x = x\,e^{t_0 x}.$$

(ii) By the mean value theorem, for each fixed x and $h \neq 0$ there is ξ between t_0 and $t_0 + h$ with $q_h(x) = x\,e^{\xi x}$. Choose $\delta > 0$ so that $[t_0-\delta, t_0+\delta] \subset \{t : M_X(t) < \infty\}$. For $|h| < \delta$,

$$|q_h(x)| \le |x|\,e^{(|t_0|+\delta)|x|}.$$

The dominating function $|x|e^{(|t_0|+\delta)|x|}$ is integrable because M_X is finite on a neighborhood of t_0 (e.g. compare with $e^{\pm(|t_0|+\delta+\varepsilon)X}$ for small $\varepsilon > 0$). Applying DCT to $E[q_h(X)]$ as $h \to 0$ yields

$$M_X'(t_0) = \lim_{h\to 0} E[q_h(X)] = E[Xe^{t_0 X}].$$

Problem 5

(a)

(i) For $X \sim \text{Laplace}(0,1)$,

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \cdot \tfrac{1}{2} e^{-|x|}\,dx.$$

For $t \in (-1,1)$ the integral splits into two convergent exponential integrals and equals $1/(1-t^2)$. For $|t| \ge 1$ one of the tails diverges. Thus

$$D_X = (-1,1).$$

In particular M_X is finite on an open neighborhood of 0.

(ii) For $X = e^Z$ with $Z \sim N(0,1)$,

$$M_X(t) = E[e^{te^Z}].$$

If $t \le 0$, then $0 < e^{te^Z} \le 1$, so $M_X(t) \le 1 < \infty$. If $t > 0$, then as $z \to +\infty$, e^{te^z} grows faster than any Gaussian decay: for large A,

$$E[e^{te^Z}] \ge \int_A^\infty e^{te^z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \infty.$$

Thus $D_X = (-\infty, 0]$, which contains no open neighborhood of 0. So the lognormal distribution does *not* have an MGF in the usual sense.

(b)

Since $Z \sim N(0,1)$,

$$E[X^k] = E[e^{kZ}] = M_Z(k) = e^{k^2/2} < \infty.$$

Finiteness of all moments does not imply finiteness of the MGF near 0: the series $\sum t^k E[X^k]/k!$ need not have a positive radius of convergence. Here $E[X^k] = e^{k^2/2}$ grows so fast that for every $t > 0$,

$$\sum_{k=0}^\infty \frac{t^k}{k!} e^{k^2/2} = \infty,$$

consistent with $M_X(t) = \infty$ for $t > 0$.

Problem 6

(a)

Write $X = \sum_{k=0}^\infty 1_{\{X>k\}}$. Taking expectations (nonnegative terms),

$$E[X] = \sum_{k=0}^\infty P(X > k) = \sum_{k=0}^\infty \big(1 - F_X(k)\big).$$

(b)

For $x \ge 0$,

$$\int_0^\infty 1_{\{X>x\}}\, dx = X,$$

so by Tonelli/Fubini,

$$E[X] = E\Big[\int_0^\infty 1_{\{X>x\}}\, dx\Big] = \int_0^\infty P(X > x)\, dx = \int_0^\infty \big(1 - F_X(x)\big)\, dx.$$

This is the continuous analogue of the discrete tail-sum formula in (a).

Problem 7

(a)

Writing $a^{\underline{m}} = a(a-1)\cdots(a-m+1)$,

$$P(X = x \mid N, M, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}} = \binom{K}{x}\frac{M^{\underline{x}}\,(N-M)^{\underline{K-x}}}{N^{\underline{K}}}.$$

Write

$$\frac{M^{\underline{x}}}{N^{\underline{x}}} = \prod_{j=0}^{x-1}\frac{M/N - j/N}{1 - j/N} \to p^x,$$

$$\frac{(N-M)^{\underline{K-x}}}{(N-x)^{\underline{K-x}}} = \prod_{j=0}^{K-x-1}\frac{1 - M/N - j/N}{1 - x/N - j/N} \to (1-p)^{K-x},$$

as $N, M \to \infty$ with $M/N \to p$ and K, x fixed. Therefore

$$P(X = x \mid N, M, K) \to \binom{K}{x} p^x (1-p)^{K-x}.$$

(b)

Using the same notation,

$$P(X = x \mid N, M, K) = \frac{K^{\underline{x}}}{x!}\cdot\frac{M^{\underline{x}}\,(N-M)^{\underline{K-x}}}{N^{\underline{K}}} = \frac{(KM/N)^x}{x!}\left(\prod_{j=0}^{x-1}\frac{1 - j/K}{1 - j/N}\right)\left(\prod_{j=0}^{x-1}(1 - j/M)\right)\frac{(N-M)^{\underline{K-x}}}{(N-x)^{\underline{K-x}}}.$$

As $N, M, K \to \infty$ with $M/N \to 0$ and $KM/N \to \lambda$ (hence also $K/N \to 0$), each finite product over $j = 0, \ldots, x-1$ tends to 1. For the remaining factor,

$$\frac{(N-M)^{\underline{K-x}}}{(N-x)^{\underline{K-x}}} = \prod_{j=0}^{K-x-1}\frac{1 - M/N - j/N}{1 - x/N - j/N} = \exp\left(\sum_{j=0}^{K-x-1}\log\frac{1 - M/N - j/N}{1 - x/N - j/N}\right).$$

For large N,

$$\log\frac{1 - M/N - j/N}{1 - x/N - j/N} = -\frac{M}{N} + O\Big(\frac{K}{N^2} + \frac{M^2}{N^2}\Big),$$

uniformly in $j \le K$, so the sum equals $-(K-x)M/N + o(1) \to -\lambda$. Therefore the product tends to $e^{-\lambda}$, and

$$P(X = x \mid N, M, K) \to \frac{e^{-\lambda}\lambda^x}{x!}.$$

Problem 8

(a)

Let $Z \sim N(0,1)$, so X has the law of $|Z|$. With the substitution $u = x^2/2$ (so $du = x\,dx$),

$$E[X] = \frac{2}{\sqrt{2\pi}} \int_0^\infty x\, e^{-x^2/2}\, dx = \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-u}\, du = \frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}},$$

and $E[X^2] = E[Z^2] = 1$. Hence

$$\mathrm{Var}(X) = 1 - \frac{2}{\pi}.$$

(b)

Let $Y = X^2$. The density of X is $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$ for $x > 0$, so for $y > 0$,

$$f_Y(y) = f(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{2}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}}\, y^{-1/2} e^{-y/2} = \frac{1}{\Gamma(1/2)\, 2^{1/2}} y^{1/2-1} e^{-y/2}.$$

Thus $Y \sim \mathrm{Gamma}(\alpha, \beta)$ with shape $\alpha = 1/2$ and scale $\beta = 2$. So $g(x) = x^2$, $\alpha = 1/2$, $\beta = 2$.

Problem 9

(a)

Assume $X \sim \mathrm{Exp}(\beta)$ has density $f_X(x) = \frac{1}{\beta} e^{-x/\beta}$ for $x > 0$. Let $Y = X^{1/\gamma}$ with $\gamma > 0$. Then $X = Y^\gamma$ and

$$f_Y(y) = f_X(y^\gamma) \cdot \gamma y^{\gamma-1} = \frac{\gamma}{\beta}\, y^{\gamma-1} \exp\bigl(-y^\gamma/\beta\bigr), \qquad y > 0.$$

This is the Weibull(γ, β) density. It integrates to 1 by the substitution $u = y^\gamma/\beta$:

$$\int_0^\infty \frac{\gamma}{\beta} y^{\gamma-1} e^{-y^\gamma/\beta}\, dy = \int_0^\infty e^{-u}\, du = 1.$$

The mean and variance are

$$E[Y] = \beta^{1/\gamma} \Gamma\Bigl(1 + \frac{1}{\gamma}\Bigr),$$

$$\mathrm{Var}(Y) = \beta^{2/\gamma} \left(\Gamma\Bigl(1 + \frac{2}{\gamma}\Bigr) - \Gamma\Bigl(1 + \frac{1}{\gamma}\Bigr)^2 \right).$$

(b)

Assume $X \sim \mathrm{Gamma}(3/2, \beta)$ has density

$$f_X(x) = \frac{1}{\Gamma(3/2)\, \beta^{3/2}} x^{1/2} e^{-x/\beta}, \qquad x > 0,$$

with $\Gamma(3/2) = \sqrt{\pi}/2$. Let $Y = (X/\beta)^{1/2}$, so $X = \beta Y^2$ and $|dx/dy| = 2\beta y$. Then for $y > 0$,

$$f_Y(y) = f_X(\beta y^2) \cdot 2\beta y = \frac{1}{\Gamma(3/2)\, \beta^{3/2}} (\beta y^2)^{1/2} e^{-y^2} \cdot 2\beta y = \frac{2}{\Gamma(3/2)} y^2 e^{-y^2} = \frac{4}{\sqrt{\pi}} y^2 e^{-y^2}.$$

This is the Maxwell density (with the parameterization induced by the problem). Integrating in polar/Gamma coordinates, or using $Y^2 \sim \text{Gamma}(3/2, 1)$,

$$\int_0^\infty f_Y(y)\, dy = 1.$$

Moreover

$$E[Y^2] = E[X/\beta] = \frac{(3/2)\beta}{\beta} = \frac{3}{2},$$

and

$$E[Y] = \int_0^\infty y \cdot \frac{4}{\sqrt{\pi}} y^2 e^{-y^2}\, dy = \frac{4}{\sqrt{\pi}} \cdot \frac{1}{2}\Gamma(2) = \frac{2}{\sqrt{\pi}},$$

so

$$\text{Var}(Y) = \frac{3}{2} - \frac{4}{\pi}.$$